

ABDUL REHMAN



Final-Year Undergraduate - Computer Science & Engineering

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EDUCATION

Examination	Degree	Institute	Year	CGPA
Graduation	B.Tech, CSE	SRM Institute of Science & Technology	2023–2027	8.6/10.0
Class XII	ICSE	City Montessori School	2022	91.75%
Class X	ICSE	City Montessori School	2020	90%

EXPERIENCE

ExamOnline | ML Engineer Intern, New Delhi
Jun'26 – Aug'26

- Designed and deployed scalable **REST APIs** for production ML inference using **Docker** and **AWS**, delivering fault-tolerant, low-latency, and cost-effective model serving.
- Built and fine-tuned **transformer-based NLP models** for automated short-answer grading and developed an adaptive question-difficulty prediction model, improving grading agreement, reducing manual review, and enabling personalized question selection.
- Built data pipelines to clean, preprocess, and version large-scale student data; partnered with senior engineers and PMs on metrics and A/B testing, driving model improvements and adoption across two exam categories.

SKILLS

- Languages & Web:** Python, C++, JavaScript, HTML, CSS, Node.js, Express.js, React.js
- Databases:** PostgreSQL, SQL, Vector Databases (Pinecone, ChromaDB, FAISS)
- Tools & Platforms:** Git, GitHub, Docker, Kubernetes, MLflow, Matplotlib, VS Code, Visual Studio
- Machine Learning:** PyTorch, Transformers, LangGraph, LangChain, LlamaIndex, Hugging Face, OpenCV, Scikit-learn, Pandas, NumPy, Neural Networks
- GenAI / LLM:** Prompt Engineering, RAG, Fine-tuning (LoRA/QLoRA), Agentic AI, OpenAI API, Anthropic API

CERTIFICATES

- Machine Learning Specialization (3-course series) — **Andrew Ng / DeepLearning.AI**
- AWS Certified Solutions Architect
- Machine Learning Certification - **IIT Kharagpur**
- Professional Generative AI Certification — **Oracle**
- Programming Languages - Python, C++
- Full Stack Web Development Bootcamp (Angela Yu)

ACHIEVEMENTS & INTERESTS

- 1st Place** at Cherry Network Hackathon for developing **BrainFile**, an AI-powered CLI tool for token-optimized codebase context generation.
- Pre-finalist (Soar-3) in **Guidewire Hackathon 2026**.
- Clubs:** Data Science Club, IEEE Club.
- Interests:** Web development, competitive programming ([LeetCode](#)),

KEY PROJECTS

AskFileAI — Local RAG-based AI File Assistant [GitHub repo](#)
Python, LangGraph, Ollama (Qwen2.5-7B, LLaVA), ChromaDB, BM25, GraphRAG, Tesseract OCR

- Built a fully offline AI assistant answering questions across **22+ file types** (docs, code, spreadsheets, images) with multi-file upload and conversation memory.
- Used advanced RAG: hybrid retrieval, LLM re-ranking, and **GraphRAG** with multi-hop entity traversal.
- Designed a modular **LangGraph pipeline** (file loaders, embeddings, ChromaDB) with a Streamlit chat UI and source attribution.

VsmoLLM — GPT-2 Built From Scratch [GitHub repo](#)
PyTorch, Hugging Face Datasets, YAML, Weights & Biases

- Built **GPT-2 from scratch in PyTorch**, including token/position embeddings, causal multi-head attention, and Transformer blocks.
- Trained on **WikiText-103** using FP16, gradient accumulation, AdamW, cosine LR scheduling, checkpointing, and GPU optimization.
- Developed training and evaluation pipelines with Hugging Face, YAML configuration, and W&B, reporting loss and perplexity.

BrainFile — Token-Optimized Codebase Context CLI [GitHub repo](#)
Python, tree-sitter, Groq API, NetworkX, Typer, tiktoken

- A Groq-powered CLI that parses codebases via **AST analysis** and generates token-optimized context files (~2K tokens vs 15K+ raw), enabling instant AI agent onboarding across **7 languages**.
- Achieved **81% accuracy** on context relevance evaluation.

RELEVANT COURSEWORK

- Data Structures and Algorithms - Object-Oriented Programming - Deep Learning - Artificial Intelligence

LEADERSHIP

- Digital Literacy Outreach - Village Community:** Taught basic computer literacy to **45+ residents**, covering hardware fundamentals, software navigation, and internet usage to help bridge the digital divide in an underserved community.
- Helped people with organization during COVID-19.
- Team Lead - ICPBSSC International Conference:** Led a team of **15 volunteers** to plan and execute event logistics for an international conference, coordinating venue setup, scheduling, and on-ground operations.

EXTRACURRICULARS

- Cricket Team - Debate Club - Ramanujan Mathematics Club
- Won first prize in painting competition
- Gaming: Counter-Strike.